

E-Commerce Sales Dashboard — Case Study

Workshop Project — TechTip24 Power BI Workshop

Tool: Power BI | Instructor: Dr. Aditi Gupta | Role: Data Analyst (workshop participant)

1. Project Overview

This dashboard was built while attending a live Power BI workshop hosted by TechTip24 and led by Dr. Aditi Gupta. Using an e-commerce sales dataset, the goal was to practice the full workflow of turning raw transactional data into a decision-ready sales dashboard — covering KPI design, category and market breakdowns, and geographic sales analysis.

2. Objective

The workshop focused on building practical Power BI skills using a realistic business scenario. The dashboard was designed to help a hypothetical e-commerce business answer:

- What are the headline sales, profit, and shipping cost figures across the business?
- Which product categories and markets generate the most sales?
- How does sales performance vary by ship mode, country, and city?

3. Methodology

Step	What I Did
Data Preparation	Loaded the e-commerce sales dataset into Power BI and shaped the fields needed for quantity, sales, profit, and shipping cost calculations.
KPI Design	Built headline cards for Sum of Quantity, Sum of Sales, Sum of Profit, and Sum of Shipping Cost to summarize overall business performance.
Category & Market Analysis	Created a pie chart of sales by product category (Technology, Furniture, Office Supplies) and a donut chart of sales by market (APAC, US, EU, LATAM, Africa, Canada).
Geographic Analysis	Built a horizontal bar chart of sales by country and a further breakdown of sales by city to identify top-performing locations.
Ship Mode Analysis	Added a trend chart of sales by ship mode (Standard Class, Second Class, First Class, Same Day) to see how shipping choice relates to sales volume.
Interactivity	Added Category and Market filter panels so the dashboard can be sliced by product line or region.

4. The Dashboard

The final dashboard combines four headline KPIs with five supporting visuals — ship mode trend, country and city breakdowns, category split, and market split — giving a full view of where e-commerce sales are coming from.

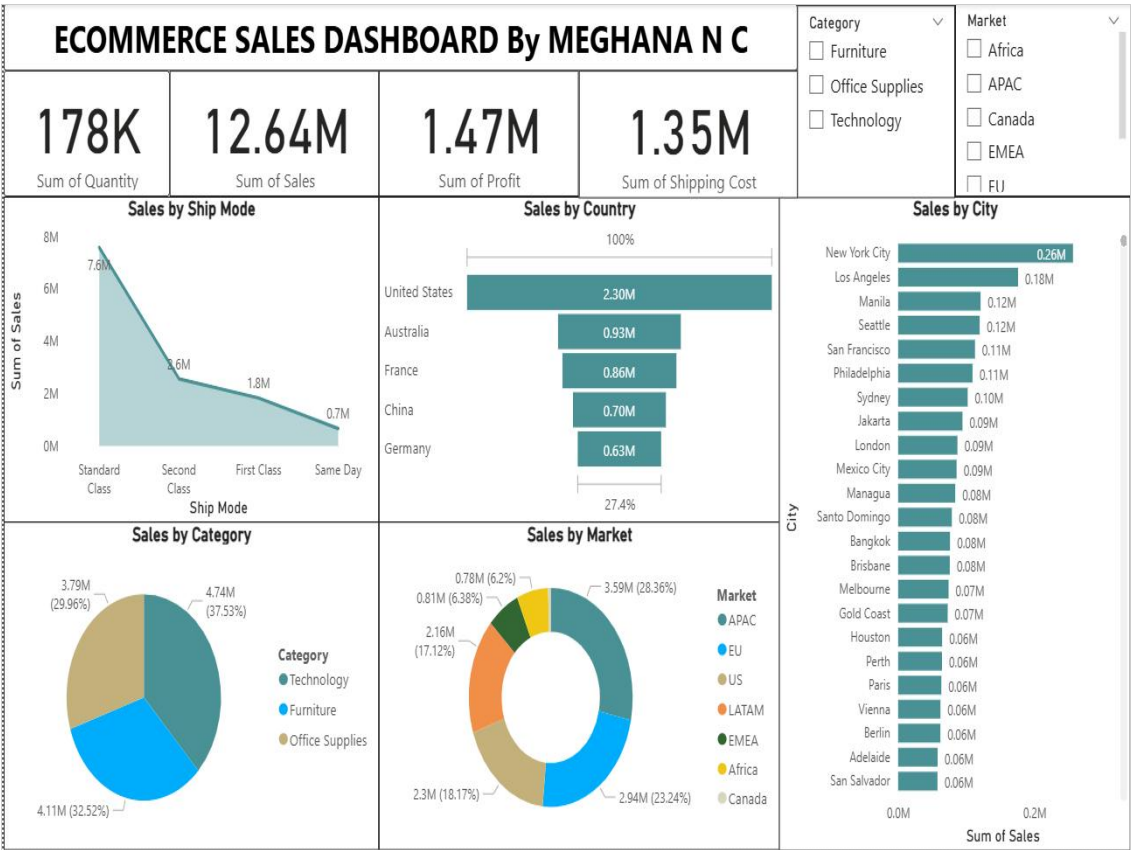


Fig. 1 — E-Commerce Sales Dashboard: KPIs, sales by ship mode, sales by country, sales by city, sales by category, and sales by market.

178K SUM OF QUANTITY	12.64M SUM OF SALES	1.47M SUM OF PROFIT	1.35M SUM OF SHIPPING COST
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5. Key Insights

- **Sales are fairly evenly split across categories:** Technology, Furniture, and Office Supplies each contribute roughly a third of total sales, suggesting a balanced product portfolio rather than dependence on a single category.
- **The US and APAC are the leading markets:** the sales-by-market donut shows the United States and APAC region generating the largest shares of sales, with EU, LATAM, Africa, and Canada contributing smaller portions.
- **Standard Class dominates shipping volume:** the sales-by-ship-mode trend shows Standard Class carrying the most sales volume, tapering down through Second Class, First Class, and Same Day — indicating most customers are not paying for expedited shipping.
- **The United States leads by country,** with Australia, France, China, and Germany following behind it in the sales-by-country breakdown.
- **New York City tops the sales-by-city ranking,** ahead of other major metros such as Los Angeles and Seattle, highlighting where demand is most concentrated.

6. Recommendations

- Continue investing across all three product categories given their balanced contribution, rather than over-indexing on one line.
- Prioritize the US and APAC markets for marketing and inventory planning, since they drive the largest share of sales.
- Investigate whether offering incentives for First Class or Same Day shipping could lift average order value, given Standard Class currently dominates.
- Use the country- and city-level views to guide regional stocking and fulfillment decisions, focusing on top cities like New York City.

7. Skills Applied

Power BI, Data Visualization, KPI Design, Dashboard Design, Data Analysis, Business Intelligence, Reporting.

8. Conclusion

Built during a hands-on Power BI workshop with TechTip24, this dashboard reflects practical, guided experience turning a raw e-commerce sales dataset into a clear, interactive business report — from KPI definition through category, market, and geographic breakdowns.